Research data: the O’Neil chain

Trunk section: §6.5 Research data.

Talks to: Sources (§6.1: SEC EDGAR, Form 13F, Polygon, Yahoo, openFDA, USASpending); the Card (§6.3), which shows the CANSLIM reading; the Market (§6.4), which shows M and L; Quant (§6.8), whose backtest gate reads the same facts as of a day.

Built on: CANSLIM, the deepest document of the old desk: the traps, the exact rules and their provenance, the card and panel specification (locked), the build phases and what each found; qp OPERATIONS (the nightly chain); CARD-FIELDS #28–#88 (what reaches the card). The code: `quant-platform/chart/oneil.py`, `relstrength.py`, `groups.py`, `sic.py`, `edgar.py`, `f13.py`, `base.py`, `ratings.py`, `canslim.py`, and `deploy/run_daily.py`.

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1. What this domain is

William O’Neil’s CAN SLIM is a method for finding leading growth stocks: **C**urrent quarterly earnings, **A**nnual earnings, **N**ew (product, management, industry conditions, price high out of a base), **S**upply and demand, **L**eader or laggard, **I**nstitutional sponsorship, **M**arket direction.

The desk reconstructs it from **free, official data**, names every reconstruction as one (`rs`, never “IBD RS”), and uses it as a **label, never a filter**: no tool’s candidates change because of it. The results are card fields and Market-tab sections; this domain computes them.

Why this domain is separate. Its facts are slow (filings, quarterly 13F, whole-market rankings), heavy (thousands of companies, rate-limited sources) and change at most once a day. They must never be computed inside a page request or a scan, and they must not depend on the chart server being up.

2. The traps the method must avoid

Most CAN SLIM concepts have a plain-English name that means something else in common use. Implementing the common reading produces numbers that look right and measure a different thing. From CANSLIM §1:

Concept	What it is	What it is mistaken for
RS Rating	a percentile 1–99 of 12-month performance against **every other stock**	RSI, which compares a stock to its own range
C	current quarter EPS vs **the same quarter a year ago**	sequential quarters (reintroduces seasonality)
A	annual EPS growth over 3–5 years, plus ROE: durability	a longer C
Distribution day	one session: index down ≥ 0.2 % on **higher volume**	any down day
Distribution count	days **live** in a rolling 25-session window, with expiry	a running total
A/D Rating	a letter A–E from 13 weeks of price and volume	Chaikin's A/D index, an unbounded cumulative line
Follow-through day	a confirmation on **day 4+** of a rally attempt, big gain on higher volume	any strong up day
N: new high	price emerging from a **base** into new ground, on volume	any 52-week high
Group rank	rank of the stock's **industry group**	the sector (far too coarse)
M	a **rule-based state machine** over two indexes	a sentiment read or an MA cross
RS line	the plotted ratio stock ÷ index; "new high **before** price" is the tell	the RS Rating
U/D volume ratio	up-day volume ÷ down-day volume over 50 days	the A/D rating
Base stage	which base counted **from the market bottom**	any consolidation
Group rank vs Group RS	**one fact**: rank 63 of 197 *is* the 68th percentile *is* B+	three measurements that could disagree
RS within the group	"RS 1 of 13": the stock ranked against its own group only	the RS Rating
N: the new thing	**three** things: product, management, **better industry conditions**	two things, and a news feed

3. The legacy method, letter by letter

Every rule below carries over. Thresholds are **O'Neil's own numbers**, named constants, printed beside any signal that fires (CANSLIM §5).

3.1 M: the market model

A state machine over the **S&P 500 (^GSPC)** and the **Nasdaq Composite (^IXIC)**, daily bars. Three published states (confirmed uptrend, uptrend under pressure, market in correction) and one internal (rally attempt).

Rule	Value	Provenance
Distribution day	close ≤ prior close × (1 − 0.002) **and** volume > prior volume **and** the market is in an uptrend	IBD's −0.2 % floor on O'Neil's plain "closes lower on higher volume" (set the floor to 0 for the plain form)
Stalling day	volume above the 25-day average or the prior day, gain under ~0.2 %, close in the lower half of the range: counts as distribution, flagged apart	IBD
Live window	25 sessions	IBD; the tight end of O'Neil's "5–6 weeks"
Removal	after 25 sessions, **or** when the index **trades** (intraday high) 5 % above that day's close	IBD
Under pressure	5 live distribution days	IBD
Correction	6+ live, or a decisive break of the rally low	IBD
The bottom	the **lowest close** of the correction (not an intraday low)	the 2010 workshop
Rally day 1	the session **after** that lowest close, up or down; a new lower close moves the bottom and restarts the count	the workshop
Follow-through	day **4 or later**, gain **≥ 1.7 %**, volume **> the prior session** (that is the whole test)	O'Neil's last figure (1 %, then 1.5 %, then 1.7 %)
Late follow-through	days 8–11 flagged late, beyond 11 very late: **flagged, never refused**	refusing creates a state the market can never leave
Status	counted **per index**, published as the **worse** of the two	"one index is enough"

Lesson recorded from the first live run. An extra test (volume above the 50-day average) blocked the Nasdaq's August 2026 follow-through; the rally ran to day 23 unconfirmed and the

model said “correction” through a confirmed rally. A rule stricter than the published one fails to the wrong answer, in the direction of telling you not to buy. The 50-day comparison is reported beside a follow-through, never decides it.

Caveat, always shown: about one follow-through in four fails. It is necessary for a bottom, not sufficient.

3.2 RS: the relative strength rating

```
RS = 0.4·(P0/P63) + 0.2·(P0/P126) + 0.2·(P0/P189) + 0.2·(P0/P252)
```

63/126/189/252 trading days = 3/6/9/12 months, the latest quarter weighted double; then **percentile-ranked 1–99 against every US stock**. Verified against IBD’s published formula. Two choices that make it match: the universe is the whole market (Polygon grouped daily, every US ticker each session), and prices are **split-adjusted** (a 2-for-1 split would otherwise score a leader at −50 %).

3.3 L: groups and the four levels of strength

Level	The number	How	Answers
Market	the M state + the index RS line	§3.1; `index ÷ S&P 500`	buy at all?
Sector	rank of a coarse sector (planned: 1 of 33)	aggregate performance	which half of the market
Group	rank X of N	**median** member RS, membership by SIC	is this the leading group (O’Neil: ~37 % of a stock’s move; sector ~12 %)
Stock in group	“RS 1 of 13”	the stock’s RS ranked inside its group	leader or laggard

Membership is the SIC code from EDGAR for every filer (TradingView labels only where SIC has none, F-21). **Thin groups** (fewer than 6 ranked members, `QP_GROUP_MIN_MEMBERS`) roll up one rung: 4-digit industry → 3-digit industry group → 2-digit major group; dropped if still thin. Groups are ranked by median RS; **percentile and letter** (A+ ≥ 94 … E) are derived from the rank, never stored as separate facts. **Rotation**: rank now against rank 3 months ago. **The divisor is stored with every rank** (IBD once went from 197 groups to 145); the card prints `63 of 197`, never a bare rank. **The honest limit**: our group rank is ours, not IBD’s, because membership differs; the rank within the group and the member count are exact.

3.4 C and A: earnings, from EDGAR

From `data.sec.gov/api/xbrl/companyfacts/CIK{…}.json` (official, free, a descriptive User-Agent; CIK ↔ ticker from `company_tickers.json`).

C: eight quarters of EPS, year-ago EPS, %Chg, sales, year-ago sales, %Chg, after-tax margin; **accelerating** (each of the last 2–3 quarters' %Chg larger than the one before); **beat +25 %** X of 8. **A**: fiscal years of EPS and %Chg (with the year's price high and low), the year **rebuilt from four quarters** to check against the filing, **3-year growth**, **stability** (the wobble around trend, 0–99, low is good), **ROE** against O'Neil's **17 %** floor.

The five EDGAR traps, each of which produces a normal-looking wrong number (CANSLIM §17.1):

#	Trap	Rule
1	C is year over year, never sequential	compare to the quarter one year back (within 45 days)
2	Q4 is usually not filed as a quarter	Q4 = FY − Q1 − Q2 − Q3, **only when all three are present**; when EPS and net income disagree in sign, Q4 is left out
3	the same period is filed more than once	the most recently **filed** value wins
4	a % from a negative base is not a number	`n/a (loss a year ago)`; capped at ±999 %
5	revenue lives under four tags	read all four, take one per period

Filings land weeks after quarter end, so **every block carries its own as-of date**.

3.5 N: the base, and what is new

The weekly base (weekly is not a display choice: a base is 7–65 weeks, and daily bars drown every shape test in noise): weeks, depth, waves down (the decline typically comes in three), **accumulation weeks** (heavy volume, no price progress), **handle** (a controlled 2–8-week drift in the **upper half** of the base: a handle that gives back half the cup is the cup failing), lip and low with their weeks, **pivot**, % away, X of 6 checks. Needs 7 weekly bars.

The RS line tell: the RS line at a new high **while price is still in its base** fires; both at new highs does not (that is arithmetic); the line off its high does not. **Divergence**: the index made a new low in the window, the stock did not, and the stock is above where it sat on the index's worst close; "the index has not made a new low" is shown as a different answer from "it followed the index down".

N, what is new (designed, phase 7, not built): see §4.

3.6 S: supply

Shares outstanding (EDGAR `dei:EntityCommonStockSharesOutstanding`, its basis when not the cover figure, as-of); **share count change over about a year**, always from the same tag (buying back / diluting / flat, or the reason there is no rate); float (TradingView) and float as % of shares; short interest and days to cover (from the card's short-interest facts); **U/D volume**

ratio over 50 sessions; **A/D**: a bounded 13-week score (−1 … +1, normalised by the volume in the window, comparable between stocks), banded to a letter that is **ours** (IBD does not publish its boundaries), with the raw number beside it.

3.7 I: institutional sponsorship, from Form 13F

The SEC's quarterly 13F datasets. Counted as distinct **managers** by CIK, unioned across every CUSIP of the issuer (stock, converts, options are one company) and across a 13F-HR and its 13F-HR/A (one manager). Per quarter: the holder count, **of N** managers who filed at all, and the **share of filers**. **Direction is read from the share**, with a 3 % relative dead band: `rising` (up over the year and still up), `peaked` (up over the year, newest quarter down), `flat`, `falling`, `turning up`; only `rising` and `turning up` are coloured bullish; the raw holder change is printed beside, uncoloured. Only managers over $100M file, 45 days after quarter end: the lag is stated on screen.

3.8 On each card: the market reflected

From CANSLIM §14, four per-card readings, each a different number on every card:

1. **Behaviour on the live distribution days**: on how many the stock's close-to-close return beat the index's, and the average difference, with the dates behind a tap; says so when there are none.
2. **Sessions since the follow-through**, banded early (< 25), established (25–150), late (> 150).
3. **Group rotation**: rank now vs 3 months ago, ▲ into / ▼ out of.
4. **The exposure implication**, in O'Neil's words.

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4. What was built, and what was not

From the build manifest (CANSLIM §12, §16):

Phase	Delivered	State
1	M: the whole market model (M1–M10)	built
2	M on the Market tab and its per-card reflection	built
3	groups: RS, Grp, volume vs 50-day, `grp X of N`, `RS X of N`, the group table, rotation, % off high	built
4	A/D, U/D ratio, RS line tell, divergence	built
5	EDGAR C and A tables, shares outstanding, the weekly base	built
6	13F I and FINRA S	I built; float estimate (S2) and funds / banks / management % (I3) not built
7	**the N pipeline**	**not built**
—	Composite (R1), EPS rating (R2), SMR (R5)	deferred: a composite without EPS and SMR would be a partial one wearing a complete name
—	group new highs / lows (L4), sector rank (L7), index RS line (M11)	not built

The N pipeline, as designed (CANSLIM §10):

```
HARVEST (24 months) → CLASSIFY (3 buckets) → REACT (price) → PERSIST (RS line) → SCORE → one line
```

Step	Rule
Harvest	EDGAR 8-K **Item 5.02** (new management: structured, the item code is the classifier), **8.01 / 7.01** (product announcements), **1.01** (material agreements); **openFDA** approvals, 510(k), PMA; **USASpending** contract awards; the group's rank rising (new industry conditions, computed). News APIs corroborate, never trigger
Classify	keep new product/service, new management, new contract, better industry conditions. **Drop** analyst notes, index changes, earnings releases (that is C and A), splits, buybacks, dividends, lawsuits, short reports. **Route** secondary offerings to S
React	on the event date + 2 sessions: the move in multiples of the stock's 20-day ATR, volume against its 50-day average. **No move beyond 1 ATR on normal volume: dropped**
Persist	the RS line today against the event day: higher 1.0, flat 0.5, lower 0.2
Score	`N_event = W(class) × R × D(age) × P`; N = the best event, 0–100. Class weights: product 1.00, management 0.90, contract 0.80, industry conditions 0.70, a new high alone 0.40. Age decay: 0–3 months 1.0, 3–6 months 0.9, 6–12 months 0.7, 12–24 months 0.4, older 0. Dedupe by class and date ± 3 sessions
Show	one line (`N 86 · New CEO · 2026-03-12 (5.8 mo) · +11.4 % on 4.2× vol · RS line held`); the event table; **unconfirmed** events (too new to have reacted) listed apart; **discarded counts printed**, as proof that it is a filter, not a feed

5. What failed

ID	Fault	Root cause	Prevented by
—	a qp restart blanks the CANSLIM reading on every card, and scans freeze empty `cs*` columns into R1	the chain lives inside the chart server, which every scan calls per card	the research worker writes facts; cards read them
Q-N5	the nightly job is a systemd timer installed by hand with a hard-coded user and path; without it the data goes stale silently	the one scheduler outside pm2	the platform scheduler runs it; a missed or failed run alerts
F-21 (fixed)	the industry map flipped between SIC and TradingView every morning	two writers for one file	one writer: the research worker
—	a missing O'Neil file made the Market tab ask qp to build it inside a page request	computation in the request path	pages only read; "not built yet · last run DATE"
—	the first live data check reported a different environment from the running one (a bare shell without the keys)	the check did not load the service's environment	health checks run inside the worker's own environment
D-D3	R2's O'Neil status read at export time	not frozen	the market state carries the M fact as it was
—	the backtest rebuilds CANSLIM "as of" a day separately	no dated facts	the gate reads facts by `as_of`
—	each piece rebuilt on first request when the job did not run, so the first person in the morning paid for it	lazy builds as a fallback	no lazy builds; the scheduler is the only trigger

6. The new design

6.1 The research worker

A Python job (it reuses qp's verified code: `relstrength.py`, `oneil.py`, `groups.py`, `edgar.py`, `f13.py`, `base.py`, `ratings.py`), started by the platform scheduler after the close and on demand. It runs its steps, writes facts, records its run, and exits. Each step is independent: one failure never stops the rest, and each failure is reported.

#	Step	Writes (fact family)	Clock	Budget
1	RS universe top-up	`bars_universe` (Polygon grouped daily, the newest sessions)	each close	minutes
2	RS ratings	`rs_rating` per ticker (with the universe size)	each close	—
3	market model	`oneil_m` for `market` (state, per-index counts, distribution days, rally, follow-through, rules in use)	each close; re-checked through the day by the market engine, labelled provisional	—
4	SIC map	`sic` per ticker	new listings and changes: nightly	—
5	groups	`group_ranking` for `market`; `oneil_l` per ticker (group, level, rank of N, percentile, letter, RS in group, rotation, roll-up)	each close	—
6	earnings	`oneil_c`, `oneil_a` per ticker	weekly, and on a new 10-Q or 10-K	a time budget per night
7	supply	`oneil_s` per ticker	monthly (shares), each close (U/D, A/D)	—
8	institutions	`oneil_i` per ticker	quarterly, after the 45-day deadline	—
9	base and RS line	`oneil_n_base`, `rs_line` per ticker	each close (weekly bars)	—
10	ratings	`oneil_ratings` (RS, Grp, A/D, volume vs 50-day; then EPS, SMR, Composite)	each close	—
11	N events	`oneil_n_events` per ticker	nightly, 24-month window	a time budget

Which tickers. Whole-market steps (1–5) cover every US stock. Per-company steps (6–11) cover the union of every ticker any tool has carded in the last 90 days, every R1 ticker, and every ticker in an open trade, so the CANSLIM reading is ready before the card asks.

6.2 Facts, with dates

Each fact family above is written to the fact store with:

- `as_of`: the date it describes (the quarter end, the session, the 13F quarter);
- `known_at`: when the worker learned it;
- `status`: `ok`, `none` (e.g. no filings) or `unavailable` (e.g. EDGAR timed out), with the reason;

- the **rules in use** for anything with a threshold, so a signal can be checked against its definition later.

This makes the backtest's CANSLIM gate exact: it reads the facts known before its decision bar (trunk §5.2).

6.3 Who reads what

Reader	Reads	Never
the card engine	`oneil_*`, `rs_rating`, `rs_line` for the card's ticker	computes any of them
the market engine	`oneil_m`, `group_ranking`	computes them (it only re-checks M against today's quote, provisionally)
quant	any of them, as of a day	writes them
pages	through the card and the market state	trigger a build

6.4 What gets built that was not

In order, each behind its own versioned definition:

1. **Index RS line** and **group new highs / lows** (cheap, from data already fetched).
2. **Sector rank** from SIC major groups, kept distinct from the group.
3. **EPS rating and SMR** from EDGAR, then **Composite** once both exist.
4. **Float estimate** (shares outstanding minus insider and 5 %-holder positions from Forms 4 and SC 13D/G), labelled as an estimate; **funds / banks / management %**.
5. **The N pipeline**, as designed in §4.

6.5 Everything explains itself

Each fact family ships with its definition card: what it means, how it is calculated, **what it is not** (from the trap table in §2), source and as-of. The card and the Market tab show it on a tap (CARD.md §5.6).

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7. Contracts

Owns (only writer)

Fact family	Subject
`bars_universe`, `group_ranking`, `oneil_m`	`market`
`rs_rating`, `rs_line`, `sic`, `oneil_l`, `oneil_c`, `oneil_a`, `oneil_s`, `oneil_i`, `oneil_n_base`, `oneil_n_events`, `oneil_ratings`	ticker

Reads

From	What
sources	EDGAR (company facts, tickers, 8-K items, SIC), 13F datasets, Polygon grouped daily, Yahoo index and stock bars, openFDA, USASpending
tools and cards	which tickers to cover
platform	its schedule and environment

Run record

Each run writes a job-run row: start, end, each step's result, counts (tickers covered, facts written, failures with reasons). A step that has not succeeded within its clock (for example, earnings not refreshed in 8 days) turns the health page amber, then red, and sends an alert.

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8. How the rebuild is proven

1. **Same numbers.** For a week the worker runs beside qp's nightly chain; for every ticker and family, its facts are compared with qp's files. Differences are explained or fixed before the card switches to the facts.
2. **qp down test.** With quant stopped, every card's CANSLIM reading and the Market tab's M and L are unchanged.
3. **The traps as tests.** Each row of §2 and each EDGAR trap of §3.4 has a fixture: a negative base, a missing Q4, a restatement, a tag switch, a split, a 13F amendment, a lowest close vs an intraday low, a day-3 bounce, a late follow-through.
4. **M against charts.** One month of distribution days and follow-throughs is checked by hand against index charts.
5. **A missed run.** Skip the worker for a night: the health page goes amber and an alert arrives; cards show "last run DATE".